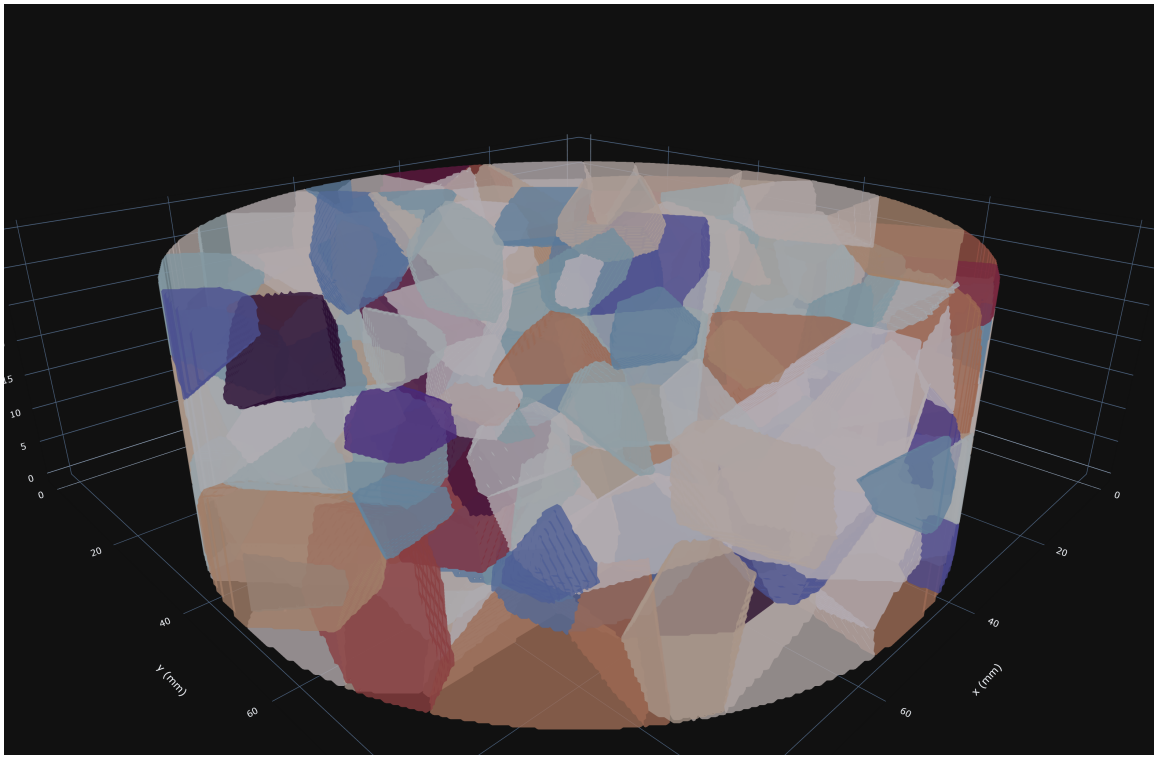
Pulse-Echo Azimuthal Simulation

Simulation Manual



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August 11, 2026

Contents

Chapter 1

Overview

1.1 What this software is

This repository is the digital twin of the pulse-echo azimuthal measurement: a rotational single-transducer ultrasound method that recovers the crystal orientation fabric (COF) of a polycrystalline ice disk from the azimuthal variation of diametral sound velocity. The physical instrument is documented in its own repository and manual; this codebase answers the questions an instrument cannot: what should the measurement see for a known fabric, which parts of the received signal carry fabric information, how do acquisition errors corrupt it, and how well can the fabric be recovered.

The pipeline has four stages, mirrored by the GUI tabs:

1. **Specimen.** Build a synthetic disk: Voronoi grain structure with controlled size distribution, and a Watson orientation distribution with controlled strength and mean direction (Chapter 2).
2. **Forward model.** Propagate pulses through the disk, either with the fast ray model (interactive) or the label-indexed anisotropic FDTD solver (full-wave, GPU), which is itself validated against an independent finite-element solver (Chapter 3).
3. **Acquisition and errors.** Simulate the rotational acquisition of the physical machine, with its error model (Chapter 4).
4. **Analysis and inversion.** Recover the fabric from the simulated traces and quantify what is and is not recoverable (Chapter 5).

The repository doubles as the evidence base for the journal manuscript on the method. The working convention throughout: every number quoted in the paper traces to a module here, and the expected output is recorded in that module's docstring. Negative results are kept deliberately, because ruling a signal path out is as much a part of the argument as ruling one in.

1.2 Quick start

1.2.1 The GUI

Double-click `Run Simulation GUI.bat` in the repository root. The first launch installs a private Python runtime into `gui/runtime/` (internet required, a few minutes) with the repository's pinned package versions plus the GUI extras; afterwards it starts offline, creates an icon-bearing shortcut, and opens at `http://localhost:8552`.

The studio walks the pipeline as a gated workflow: Specimen & Fabric, Probe & Excitation, Error model, then Acquisition; the Backscatter and COF reconstruction tabs unlock once acquired data exists. The FW sweep tab drives the resumable full-wave azimuth sweep engine.

Everything interactive (specimen builder, ray forward model, error model, reconstruction views) runs on CPU. New full-wave runs need CUDA and cupy in the running Python environment; the GUI detects the backend by inspection and reports it under the title.

1.2.2 Reproducing a stored result

No GPU is needed to reproduce any analysis in the paper. For example, the finite-element cross-check of direct-arrival attenuation:

```
cd sim/analysis
python fe_direct_attenuation.py
```

prints a table of the five cross-check rounds; the expected values, including the 0.027 dB solver agreement for the 6 to 15 mm placement, are recorded in the module docstring. Every cited module follows the same pattern: run it from its own directory, compare the printed numbers with the docstring.

1.3 The reproduction contract

The version pins in `requirements.txt` are deliberate and strict: numpy and scipy are pinned to the exact versions under which the stored results were produced. The analysis layer contains bit-exactness gates that compare recomputed arrays against archived ones at zero tolerance. Other library versions move floating point at the 10^{-13} level, which is physically meaningless but trips those gates by design; the gates exist precisely to detect that anything at all has changed. With the pinned versions every gate returns zero; with other versions, expect agreement at the 10^{-12} level and two modules stopping at their gate on purpose.

Large artefacts follow a strict rule: no bulky binary ever enters git history. Solver meshes and sweep outputs are regenerable and live outside version control; the tracked exceptions are a handful of kilobyte-scale trace archives that are the only raw evidence behind specific paper claims, kept so those claims can be checked without rebuilding a multi-hundred-megabyte mesh. `DATA_MANIFEST.md` records what data lives where.

Chapter 2

Specimen and fabric model

2.1 Grain structure

A specimen is a disk (default 100 mm diameter, 35 mm thick) filled with a Voronoi tessellation of grains. The builder (`sim/core/specimen.py`, `DiskSpecimen`) controls grain count, the spread of grain sizes about the mean (a coefficient of variation on the seed weighting), and the random seed, so specimens are exactly repeatable. The tessellation is rasterised onto the solver grid as a label field: every cell carries the integer index of its grain, and all per-grain properties (orientation, stiffness) are looked up through that label. This label-indexed representation is what makes the anisotropic solver practical: the stiffness tensor field never exists in memory, only labels and a per-grain property table.

The builder also reports the diagnostics that matter physically: achieved grain count and mean equivalent diameter, the orientation tensor eigenvalues of the realised fabric, and the mean disorientation across grain boundaries, which is the quantity that drives scattering strength. The title page shows a 3D rendering of a 300-grain specimen with each grain coloured by its in-plane c-axis azimuth; Figure 2.1 shows the same specimen's mid-plane slice.

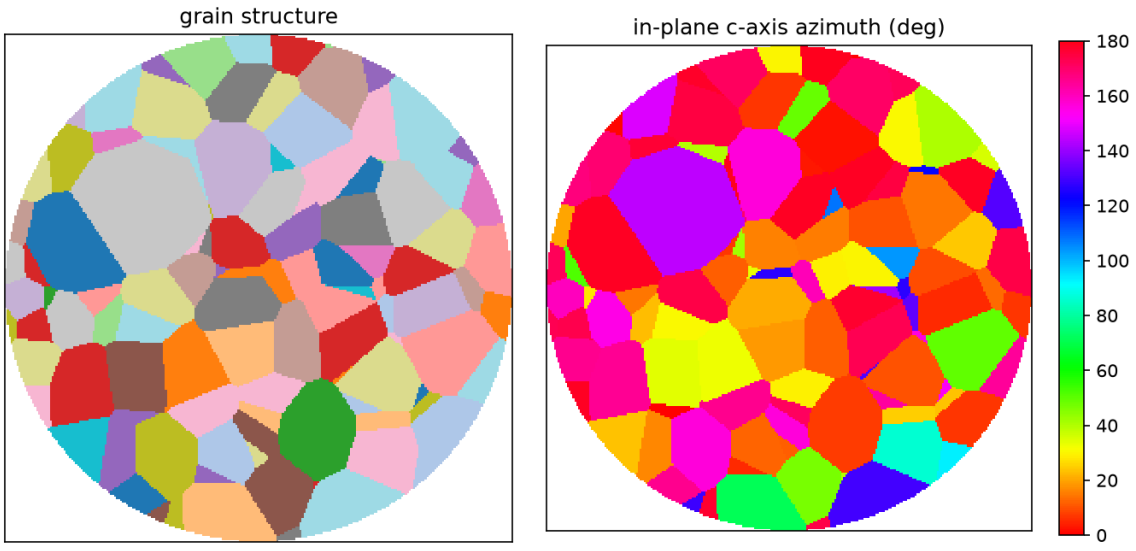

Figure 2.1: Mid-plane slice of a 300-grain specimen: grain structure (left) and the in-plane c-axis azimuth field (right).

2.2 Crystal orientation fabric

Each grain's c-axis is drawn from a Watson distribution about a chosen mean direction. The user-facing strength parameter is the leading orientation-tensor eigenvalue a_1, the number ice thin sections actually report: $a_1 = 1/3$ is random, $a_1 = 1$ a perfect single maximum. The builder converts a_1 to the Watson concentration κ internally (`kappa_for_eigenvalue`), so simulated fabrics are specified in the same units as the field data they are compared against.

The mean direction is selectable. The default is in-plane, because the azimuthal method reads the c-axis through second-harmonic velocity patterns that an in-plane fabric produces directly; a vertical (disk-normal) fabric is the method's null case and is available precisely to demonstrate that. A spatial-correlation control makes neighbouring grains more alike, which lowers boundary disorientation and therefore scattering, independent of the bulk fabric strength: bulk anisotropy and local disorder are deliberately separate knobs, because the first drives the coherent 2φ velocity signal while the second drives the incoherent scattered field.

Grain orientations, like the tessellation, are seed-repeatable, and the realised (not target) orientation tensor is always reported, since a 100-grain specimen realises the target distribution only approximately. Figure 2.2 shows the realised fabric of the title-page specimen on an equal-area pole figure.

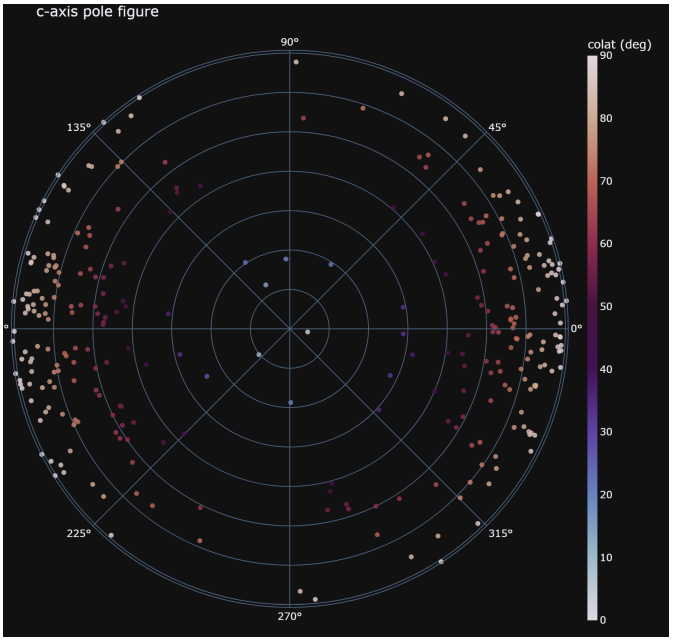

Figure 2.2: Equal-area (Schmidt) lower-hemisphere pole figure of the 300-grain specimen at $a_1 = 0.70$ with an in-plane mean c-axis: a moderate single-maximum fabric.

Chapter 3

Forward model and validation

3.1 Solvers

Three forward models coexist, used at different costs for different questions:

Ray model (`sim/model/`, used by the studio). Pure numpy, interactive speed. Propagates diametral transits through the per-grain slowness field to predict travel times and the azimuthal velocity pattern. This is the model the inversion inverts, and the GUI's default backend.

Anisotropic FDTD (`sim/core/fdtd.py`). The full-wave workhorse: a label-indexed elastic finite-difference solver in which every cell's stiffness comes from its grain's orientation. Runs on GPU through cupy; produces the synthetic pulse-echo traces the analysis layer consumes. A pseudospectral solver provides an independent numerical scheme for validation.

Finite-element arbiter (`sim/fe_crosscheck/`). A completely independent solver stack (its own mesher, elements and probes) used to adjudicate the FDTD results. When two unrelated discretisations of the same physics agree, the result is a property of the physics rather than the scheme.

The rotation machinery rotates the specimen rather than the probe, matching the physical machine, and the bit-equivalence gate (`validate_labels.py`) guarantees the rotated label field feeds the solver exactly the specimen the builder produced.

3.2 Discretisation rules

The production discretisation was fixed by convergence study, not convention, and the settled rules are enforced as defaults:

- at least 6 grid cells per wavelength at the source centre frequency,
- eighth-order spatial finite differences,
- boundary damping factor 0.02 in the absorbing layers.

Runs below these settings show visible numerical dispersion in the coda and are suitable only for previews (the GUI's specimen preview deliberately rasterises coarser, and says so). One historical trap is documented for future users: early reference traces generated before the absorbing-boundary settings were finalised carried a boundary artefact that contaminated apparent scattering floors; the clean reference set in `sim/ref/` postdates that fix, and the analysis modules that measured the difference are kept as evidence.

3.3 The validation chain

The solver's license to be believed is a chain of checks, each stored with its expected numbers in the corresponding module docstring:

1. **Analytic checks** (`sim/validation/`): rotated staggered-grid behaviour and exact isotropic reflection coefficients against closed-form results.
2. **Internal cross-scheme checks** (`sim/fw_checks/`): the FDTD solver against the pseudospectral solver on shared problems, oblique-incidence validation, anisotropy checks and convergence studies, plus the crop and corridor licenses that justify the production domain sizes.
3. **The independent arbiter** (`sim/fe_crosscheck/`): five rounds of cross-checks against the finite-element stack. The headline result, reproducible from the archived traces without a GPU, is direct-arrival attenuation agreement between the two solvers to within hundredths of a decibel (0.014 to 0.068 dB across the valid rounds).

The chain is ordered by independence: each later stage shares less with the solver under test than the one before. Chapter 5 leans on this: statements about what the scattered field does and does not contain are only as good as the solver producing it. Figure 3.1 shows the final arbiter round: the grain-scattered field isolated by the same-specimen difference in each solver, tracking between two discretisations that share nothing but the physics.

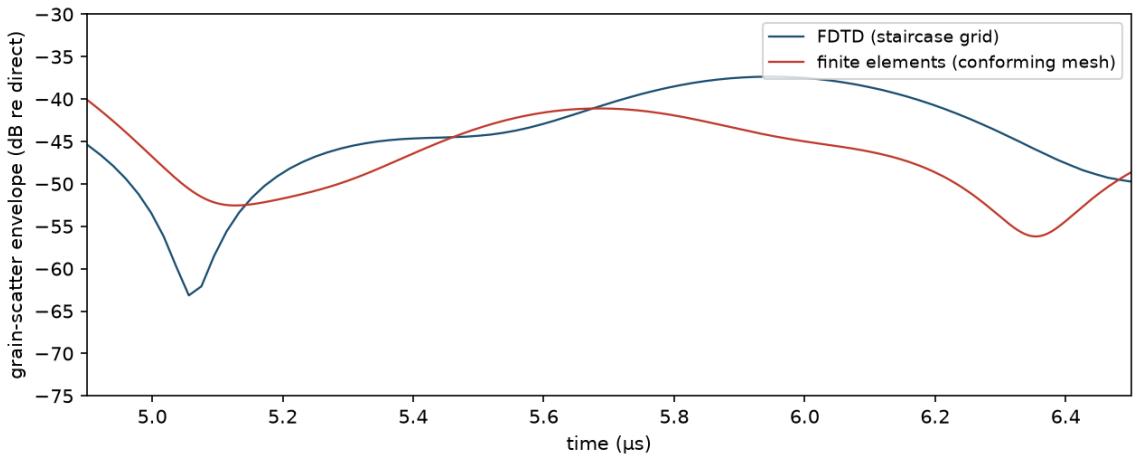

Figure 3.1: The passing arbiter round, reproduced from the archived traces: grain-scatter envelopes (specimen minus uniform control, each referenced to its own direct arrival) for the FDTD solver and the independent finite-element solver over the analysis window.

Chapter 4

Acquisition and the error model

4.1 Probe and excitation

The simulated probe mirrors the physical instrument: a single finite-aperture transducer on the disk rim, firing a broadband pulse across the diameter and receiving its own echoes. The Probe & Excitation tab sets aperture, centre frequency, bandwidth and pulse shape; the source is injected with the aperture's true footprint so diffraction and beam spread are represented rather than assumed.

Development followed the physical campaign at 2 MHz: the working rule for this project is to perfect the method at 2 MHz before spending anything on the 5 MHz production sweeps, because every signal-path question so far has been resolvable at the lower frequency for a fraction of the compute.

4.2 The sweep engine

Full-wave azimuth sweeps are driven by `sim/pipeline/sweep_runner.py`, exposed in the GUI's FW sweep tab. A sweep rotates the specimen through a set of azimuths and runs the solver at each, exactly as the machine rotates the disk under its probe. Sessions are resumable by construction: each sweep directory under `out/sweeps/` holds a `config.json` and one trace archive per completed azimuth, so an interrupted campaign continues where it stopped, and the analysis layer reads completed azimuths without waiting for the rest.

Each azimuth's traces are stored raw. Everything derived (envelopes, picks, fits, backprojections) is recomputed downstream, so a stored sweep never bakes in an analysis decision. Figure 4.1 shows the anatomy of one stored trace.

The arc backprojection view (`sim/pipeline/`) maps coherent energy in the received traces back into the disk and is the quickest qualitative check that a sweep contains structure rather than noise.

4.3 The error model

The error model corrupts ideal simulated acquisitions with the distortions the physical machine actually exhibits, so inversion robustness is tested against the right enemies rather than generic noise. The configurable terms mirror the instrument's error budget: timing jitter, per-dock coupling variation (the re-seated couplant film), angular error on the commanded azimuth, slow drift across a sweep (the thermal term the machine corrects with reference revisits), and additive noise at the receiver.

Because the same fabric can be re-acquired with different error draws, the pipeline separates what limits recovery: fabric ambiguity (Chapter 5), acquisition corruption, or both. The studio's

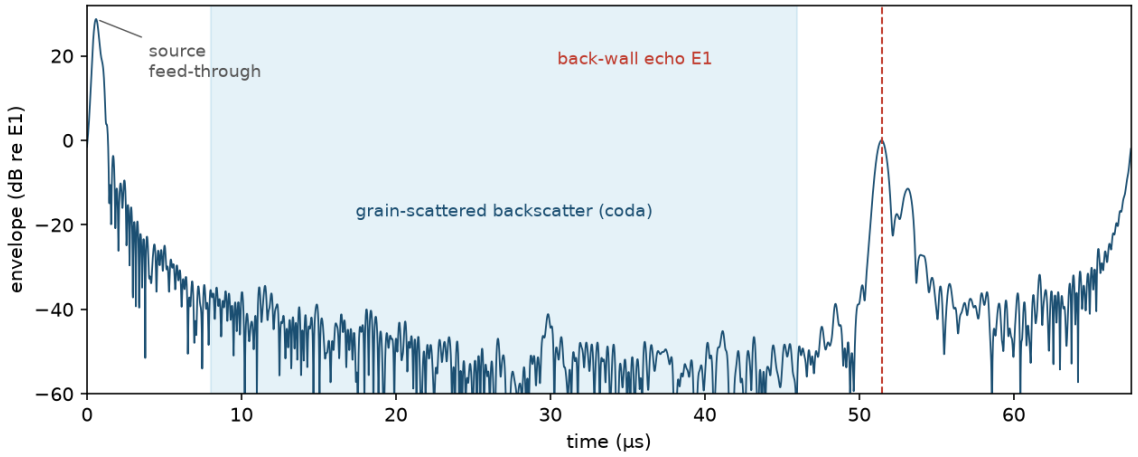

Figure 4.1: A clean 2 MHz reference trace (aligned fabric, envelope in dB relative to E1): source feed-through, the grain-scattered backscatter that fills the record, and the back-wall echo E1 near 52 microseconds whose time of flight carries the velocity observable.

gated workflow enforces the order: an error model is chosen before acquisition, and the acquired dataset records the model that produced it.

Chapter 5

Analysis and inversion

5.1 Fabric inversion

The primary observable is the azimuthal velocity pattern $v(\varphi)$, read through the E1 time of flight (Figure 5.1), and the primary inversion (`sim/model/fit_fabric.py`, `fit_sweep.py`) recovers fabric parameters from it in two stages: a coarse global search over fabric strength and direction followed by local refinement, which avoids the local minima that a direct descent on the periodic 2φ pattern invites. The fit reports the recovered eigenvalue and fast-axis azimuth with residuals, and the sweep-level tooling fits entire simulated campaigns unattended.

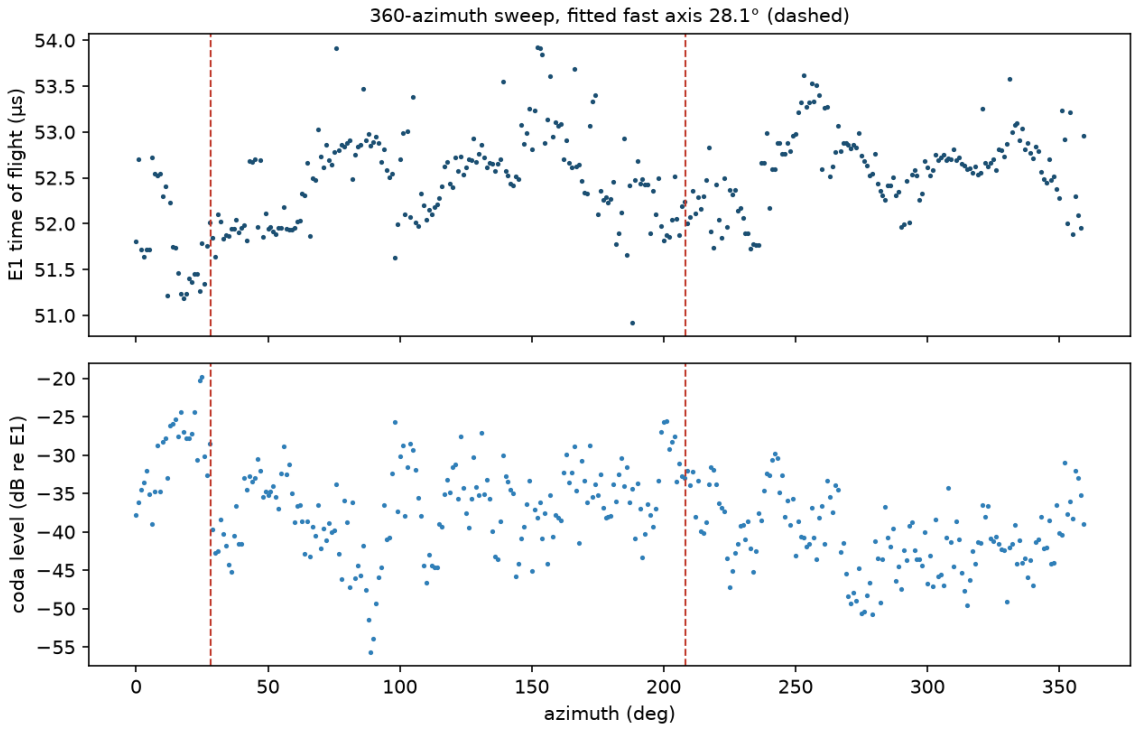

Figure 5.1: A complete 360-azimuth full-wave sweep: E1 time of flight (top) and coda level (bottom) against azimuth. The second-harmonic time-of-flight pattern and the coda elevation around the fabric axis are both visible by eye; dashed lines mark the fitted fast axis.

Beyond the single bulk fit, a gridded inversion divides the disk into regions and inverts per-area fabric, with a daemon that processes sweeps as they complete. Its purpose is honesty about spatial resolution: a diametral measurement integrates along chords, so per-area recovery is only meaningfully constrained where chords cross, and the gridded results quantify that rather than

assert it.

5.2 The scattered field

A large fraction of the analysis layer interrogates the incoherent part of the received signal: the grain-scattered coda that arrives after the specular echoes. The central question was whether the coda carries usable fabric information of its own.

The settled model is geometric: the late field is described by contrast-weighted facet reflections, in which each grain-boundary facet contributes according to its orientation and the acoustic contrast across it. The model was accepted only after it survived three independent controls (its supporting modules and their expected numbers live under `sim/analysis/`, primarily the facet-model and physical-optics themes), and it explains the observed coda statistics without invoking fabric-specific propagation effects.

That last clause is the important one, and it is a negative result: see the next section.

5.3 Negative results are kept

The repository retains every serious attempt that failed, with the measurements that killed it, because the paper's claims about what the measurement can do rest equally on demonstrations of what it cannot.

The flagship negative result concerns the coda: multiple independent routes (correlation structure, frequency behaviour, channel statistics, cross-fabric comparisons) each failed to extract fabric information from the scattered tail beyond what the coherent 2φ velocity pattern already provides. The conclusion, tested from four directions before being accepted, is that on this geometry the coda does not measure fabric; what survives is the facet-scattering description of the previous section. The analysis modules behind each route remain runnable, docstrings recording what they were expected to show and what they showed instead.

Practical guidance for anyone extending the work: before pursuing a new signal path, read `sim/analysis/README.md` and the negative results first. The most tempting ideas have usually been tried, and the stored evidence says exactly how they died.

Chapter 6

Software

6.1 Repository architecture

The code is organised by role, and `sim/README.md` is the authoritative file-by-file map; this section gives the shape.

sim/core/ the solver and specimen machinery: the label-indexed anisotropic FDTD solver, the pseudospectral solver, the specimen builder, rotation machinery, and the bit-equivalence gate.

sim/model/ forward models and inversion: the born ladder, the fabric and sweep fits, the gridded inversion and its daemon, and the ray model the studio uses.

sim/pipeline/ drivers: the resumable sweep engine, trace lab, arc backprojection and visualisation.

sim/fe_crosscheck/ the independent finite-element arbiter and its archived cross-check rounds.

sim/fw_checks/ and sim/validation/ the validation chain of Chapter 3.

sim/analysis/ roughly 75 analysis modules with their stored results, grouped by theme, each self-contained and runnable from its own directory.

sim/ref/, sim/runs/, sim/results/, sim/figures/ reference traces, batch campaign drivers, stored campaign results, and the paper's figure scripts.

gui/ the Streamlit studio, the sweep tab, and the launcher runtime scripts.

vendor/ vendored dependencies, so the repository is self-contained.

Two conventions hold everywhere: modules read stored inputs relative to their own location, so they run from their own directory with no environment setup; and expected outputs are recorded in docstrings, so checking a module means running it and comparing.

6.2 The studio GUI

The GUI (`gui/app.py`, Streamlit) presents the pipeline as a gated workflow across tabs: Specimen & Fabric, Probe & Excitation, Error model, Acquisition, then Backscatter and COF reconstruction, the last two unlocking only once acquired data exists in the session. The FW sweep tab (`gui/sweep_tab.py`) configures, launches and monitors full-wave sweeps through the sweep engine.

The specimen views are worth knowing about even for command-line users, since they are the fastest way to sanity-check a fabric: a 3D per-grain isosurface rendering coloured by c-axis, a c-axis vector field, an equal-area pole figure with orientation-tensor eigenvalues, and 2D slices of the grain and azimuth fields.

One engineering note recorded here because it is invisible in the code's behaviour and expensive to rediscover: the GUI never probes the GPU at import time. Streamlit executes the script in a worker thread, and initialising CUDA off the main thread crashes the whole server without a traceback; the backend is therefore detected by inspection (is cupy importable, is a CUDA path configured) and the solvers touch the driver only when actually run.

6.3 Data layout and environments

6.3.1 Where data lives

out/sweeps/ one directory per sweep session: a `config.json`, one trace archive per completed azimuth, and fit results. Resumable and analysable while incomplete.

out/tesscache/, out/observables/ the tessellation cache and observable matrices, regenerated on demand.

sim/analysis/*/ (stored npz) each analysis theme stores its own results beside its code.

sim/ref/ the clean reference traces behind the paper's figures.

None of `out/` is version-controlled; it is deposited with the paper. `DATA_MANIFEST.md` records the mapping. The large finite-element meshes are not versioned; the meshing scripts rebuild them if rerun.

6.3.2 Environments

Three usage tiers, in increasing weight:

1. **Analysis only** (no GPU): the pinned numpy and scipy of `requirements.txt`. Every stored result reproduces on CPU.
2. **GUI**: the pinned set plus streamlit and plotly (matplotlib is already required for figures). The launcher's private runtime installs exactly this.
3. **New solver runs**: additionally CUDA and cupy. Only needed to regenerate sweeps or reference traces.

6.3.3 The manual

This document builds from `docs/` with `build.ps1` (XeLaTeX, two passes). Each chapter is a numbered folder and each section a separate file, so updates touch a single small file.

Related publications

- J. Graves, S. Harput, B. Lishman. *A Finite-Difference Simulation Framework for Ultrasonic Crystal Orientation Fabric Estimation in Ice: Timing Methodology, Forward Model Selection, and the Cramér-Rao Accuracy Limit.* IEEE International Ultrasonics Symposium, 2026 (accepted). The paper this repository underpins.
- J. Graves, S. Harput, B. Lishman. *Non-Destructive Ultrasonic Estimation of Ice Crystal Orientation Fabric: Multi-Frequency Experimental Validation and Failure Mode Characterisation.* IEEE International Ultrasonics Symposium, 2026 (accepted). The experimental companion, measured on the physical scanner.
- J. Graves, B. Lishman, S. Harput. *Measuring Predominant Orientations of Ice Crystal Fabrics From Ultrasonic Measurements of Ice Cores.* Preprint.
- J. Graves, B. Lishman, S. Harput. *Determining the Grain Geometry From Ultrasonic Measurements of Large-Grained Temperate Ice Cores.* IEEE International Ultrasonics Symposium, 2023. doi:10.1109/ius51837.2023.10307539.

The full list is maintained at jeromegraves.com.